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FAULT 2

Probability Reversal

What is evidence for what?

The Two Questions

Two probabilities that sound the same and are not

WHAT CAN BE MEASURED

P(observation | condition)

When the condition is there, how often do we see this?

WHAT YOU WANT TO KNOW

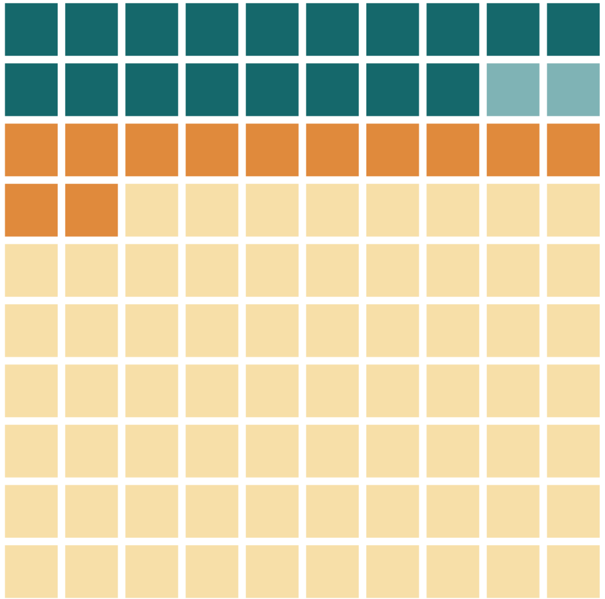
P(condition | observation)

Having seen it, how likely is the condition?

The fault is reading one as if it were the other. They are different numbers, and how far apart they sit depends on something the observation never tells you: how common the condition is.

The two questions, in one picture

Same 18 on top



P(wet | rained)

18 ÷ 20 rainy mornings = 90%

P(rained | wet)

18 ÷ 30 wet mornings = 60%

The top number never changed.

Only the group you divide by.

Each square is one morning. Every probability here is one count divided by another: 18/20 is P(lawn wet | rain), 18/30 is P(rain | lawn wet). Two minutes.

Every test reports two numbers, and each one flips

SENSITIVITY

P(test positive | have the cause)

what the test is measured on

↓ **reversed**

P(have the cause | test positive)

SPECIFICITY

P(test negative | no cause)

what the test is measured on

↓ **reversed**

P(no cause | test negative)

Each reversal is undone by the other number. A positive is undone by the false positives. A negative is undone by the false negatives.

A Sensitive test with a Negative result rules out. A Specific test with a Positive result rules in.

THE TWO EXAMPLES

Where each reversal does damage

A SCREENING MAMMOGRAM

Good at finding the cancers that are there.

high sensitivity

WHERE IT MISLEADS

A positive.

A COVID RAPID TEST

Good at clearing the people who are well.

high specificity

WHERE IT MISLEADS

A negative.

Every test has both reversals. We take each one where it does more damage.

The picture can't be displayed.

THE FIRST REVERSAL

A Screening Mammogram

If it is positive, do I have cancer?

A REAL TEST, REAL NUMBERS

What the mammogram is measured on

SENSITIVITY

P(called back | cancer)

87 of every 100 women with cancer

the other 13 are missed

A positive mammogram is a **callback**. The radiologist wants more pictures. It is not a diagnosis, and nothing has been found yet.

This is the number the mammogram is judged on, and it is a good one. But notice where it starts: from already knowing she has cancer. That is the only direction a test can be measured in.

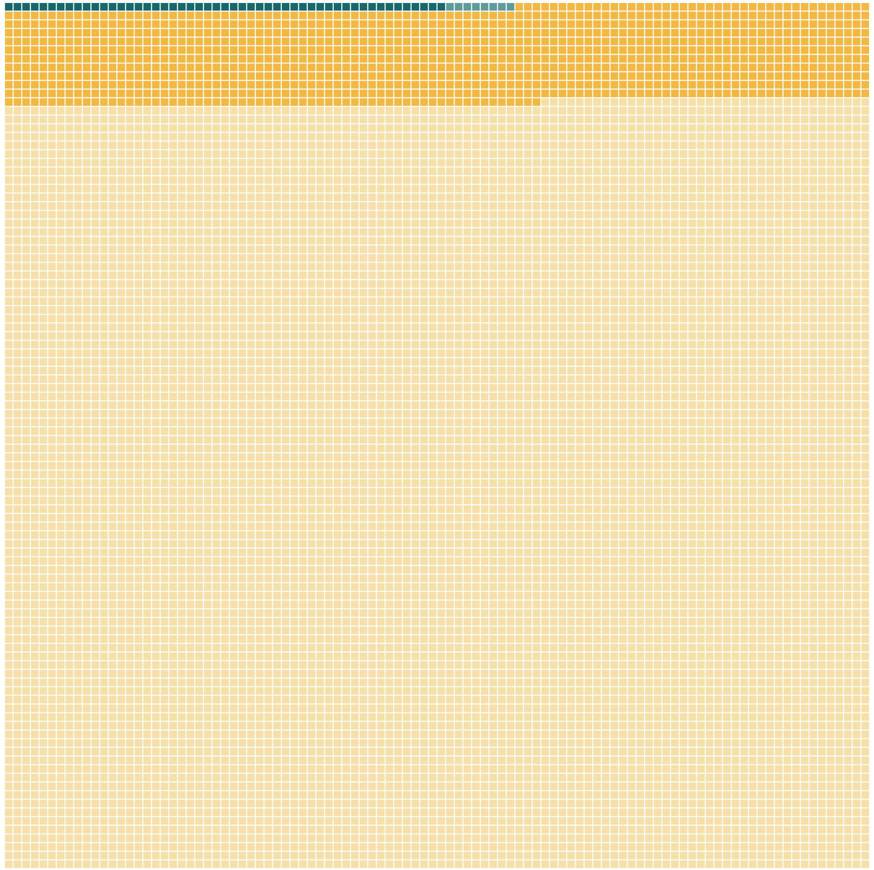
What we want is P(cancer | called back): 4.4%

Same test, same day. Now we walk the reversal.

Sensitivity 86.9%. Lehman CD et al. National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 2007-2013.

AN ILLUSTRATIVE COHORT OF 10,000 WOMEN

Six in a thousand have it



Every square is one woman screened once. The teal sliver at the top is everyone who has cancer.

That is an average across ages 40 to 89. It climbs with age: nearer 4 per 1,000 in the fifties, 6 or 7 in the sixties.

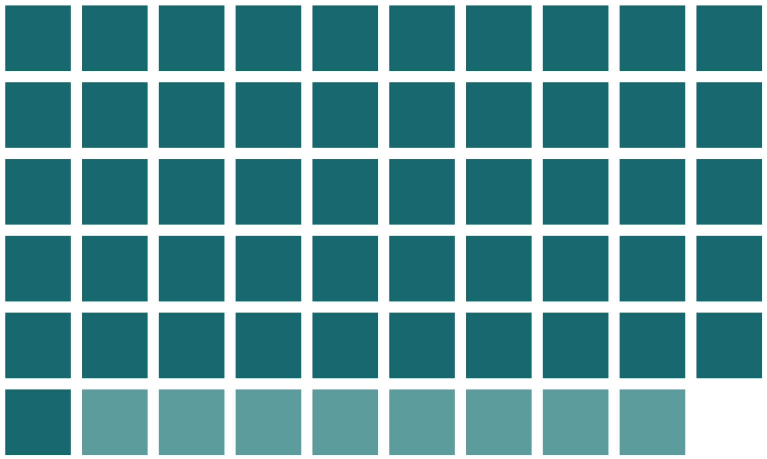
- has cancer, called back: **51**
- has cancer, not called back: **8**
- no cancer, called back anyway: **1,103**
- no cancer, all clear: **8,838**

The amber band is eleven rows deep. It is more than twenty times the size of the group the test is looking for.

BCSC benchmarks applied to a 10,000-woman cohort. Lehman CD et al. National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 359 radiologists, 95 facilities, 2007-2013.

FIRST QUESTION

If you have cancer, how likely is a callback?



The 59 women who have cancer, enlarged from the 10,000.

Look only at the 59 women who have cancer. Everyone else is set aside for now.

Of those 59, the ones called back: **51**

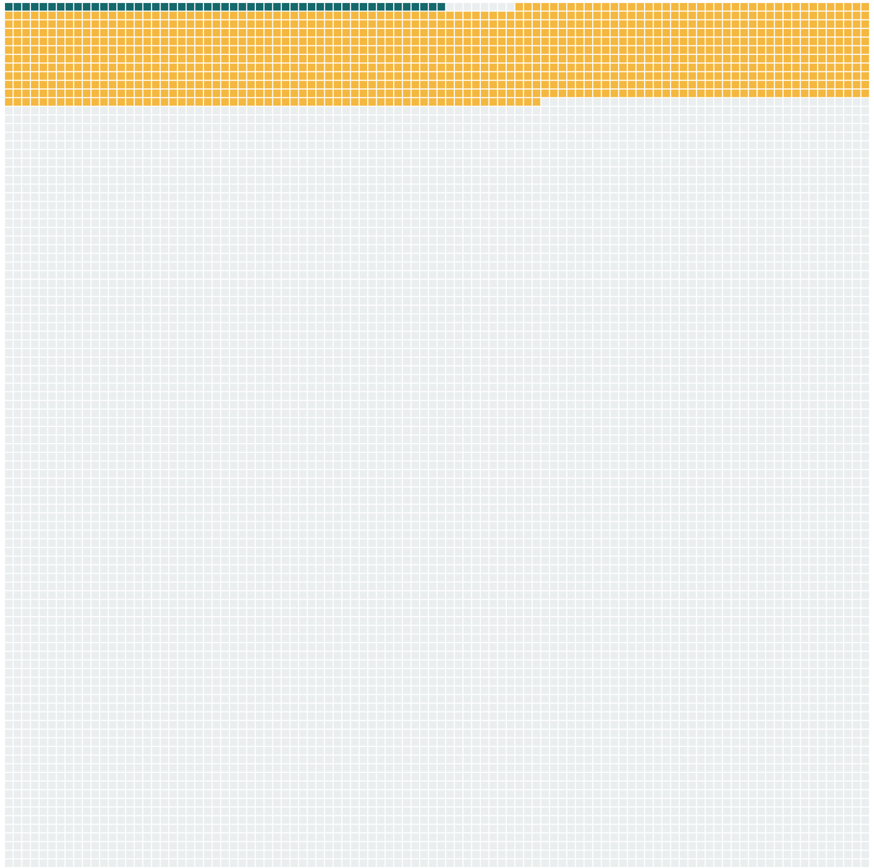
P(callback | cancer) = 51 / 59 = 87%

This is the number the mammogram is judged on, and it is a good one.

Sensitivity 86.9%. Lehman CD et al. National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 359 radiologists, 95 facilities, 2007-2013.

SECOND QUESTION, THE ONE PEOPLE GET BACKWARDS

If you get a callback, how likely is cancer?



Now look only at the women who get called back. Same picture, we just divide by a different group.

Of those 1,154, the ones who have cancer: **51**

P(cancer | callback) = 51 / 1,154 = 4.4%

Twenty-two of every twenty-three callbacks are false alarms. This is the value the benchmark study itself publishes.

Positive predictive value of an abnormal interpretation, 4.4%. Lehman CD et al. National performance benchmarks for modern screening digital mammography, Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58. 1,682,504 screens, 359 radiologists, 95 facilities, 2007-2013.

Unlikely, but not ignorable

$$P(\text{ called back } \mid \text{ cancer }) = 87\%$$

↓ *reversed becomes*

$$P(\text{ cancer } \mid \text{ called back }) = 4\%$$

87% is what the mammogram was measured on. 4% is what her letter means. Same test, same day, same woman.

Unlikely is why she should not panic. Not ignorable is why the biopsy is warranted.

Positive predictive value of an abnormal interpretation 4.4%, of a biopsy recommendation 25.6%. Lehman CD et al., Breast Cancer Surveillance Consortium. Radiology 2017;283(1):49-58.

THE BIGGER PICTURE

Why reasonable people disagree

Of US women aged 50, screened every year for ten years:

avoid a breast cancer death	**0.03% to 0.32%**
have at least one false alarm	**49% to 67%**
are overdiagnosed, and treated for it	**0.3% to 1.4%**

Nobody in this argument disputes the sensitivity, the specificity, or the 4.4%. The disagreement is over how to weigh a false alarm for half of them against a death averted for a fraction of one percent.

So the guidance splits. The American College of Radiology recommends annual screening from 40. The Task Force says every two years from 40. The Nordic Cochrane group and the Swiss Medical Board have argued for not running screening programmes at all.

Ranges from published estimates for women aged 50 screened annually for a decade. Overdiagnosis is contested: the Canadian National Breast Screening Study's 25-year follow-up put 106 of 484 screen-detected cancers (21.9%) as overdiagnoses; Gotzsche and Jorgensen, Cochrane 2013 (CD001877), found no effect on overall mortality across ten trials. An 11.6% callback rate compounded over ten independent screens gives about 71%, near the top of the false-alarm range.

THE SECOND REVERSAL

A COVID Rapid Test

If it is negative, am I clear?

A REAL TEST, REAL NUMBERS

What the swab is measured on

SPECIFICITY

P(negative | no COVID)

99 of every 100 people who are well

almost nobody healthy is wrongly flagged

That is about as specific as any test in common use. It is why a positive swab is worth believing.

But look where it starts: from already knowing the person is well. Nobody handed a negative result knows that.

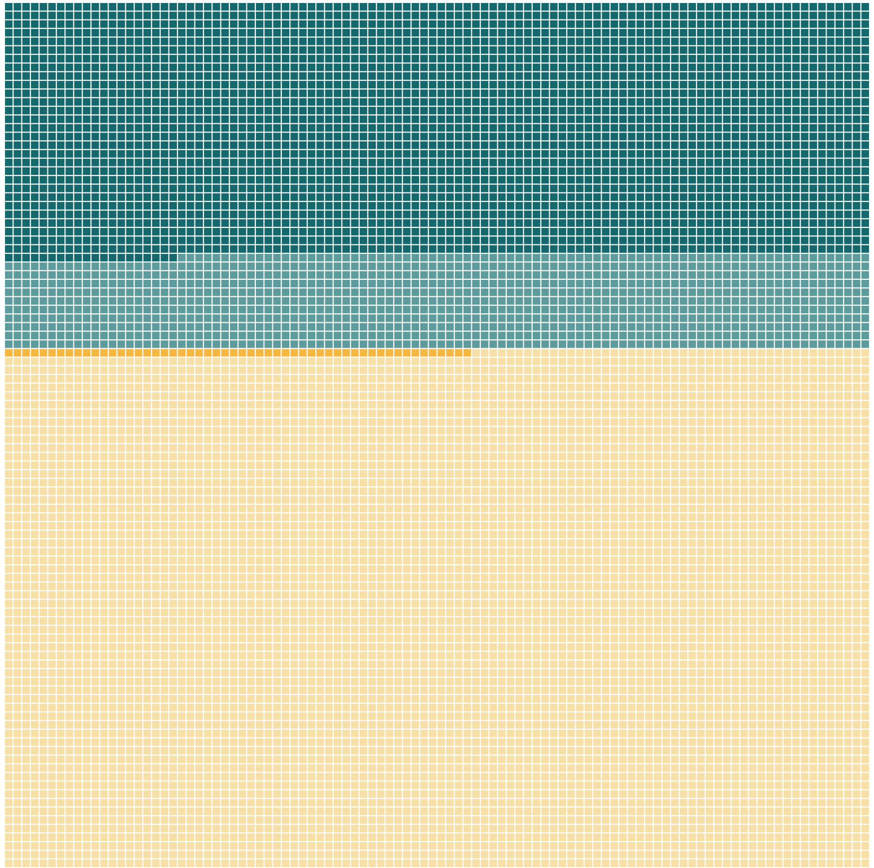
What we want is P(no COVID | negative): 85%

Nowhere near certainty. Now we walk the reversal.

Rapid antigen tests in symptomatic people: specificity 99.1%, sensitivity 73.0% (95% CI 69.3 to 76.4). Dinnes et al., Cochrane Database of Systematic Reviews 2022;7:CD013705.

AN ILLUSTRATIVE COHORT OF 10,000 PEOPLE WITH SYMPTOMS

Four in ten of them have it



Everyone here feels ill during a wave, so the odds before testing are far higher than in the street outside.

- has COVID, tests positive: **2,920**
- has COVID, tests negative: **1,080**
- no COVID, tests positive: **54**
- no COVID, tests negative: **5,946**

The pale teal band is the one to watch. Eleven rows of people who have COVID and were told they did not.

Pre-test probability of 4 in 10 is illustrative, chosen to represent symptomatic testing during a wave. Test figures: Rapid antigen tests in symptomatic people: sensitivity 73.0% (95% CI 69.3 to 76.4), specificity 99.1%. Dinnes et al., Cochrane Database of Systematic Reviews 2022;7:CD013705.

THE REVERSAL, FOR ONE PERSON

A negative is not an all-clear

$$P(\text{ negative } | \text{ no COVID }) = 99\%$$

↓ *reversed becomes*

$$P(\text{ no COVID } | \text{ negative }) = 85\%$$

99% is what the swab was measured on. 85% is what your negative means.

More than one person in seven who was told they were negative had COVID and went out anyway. A negative narrows the odds. It does not clear you.

Counts carried from the previous slide: 7,026 negatives, of which 1,080 had COVID. Rapid antigen tests in symptomatic people: specificity 99.1%, sensitivity 73.0% (95% CI 69.3 to 76.4). Dinnes et al., Cochrane Database of Systematic Reviews 2022;7:CD013705.

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WHEN NOBODY CATCHES IT

Courtrooms and Classrooms

What did the flip cost?

A CAUTIONARY TALE

When the reversal reaches a verdict

A RAPE TRIAL IN ENGLAND, CONVICTION QUASHED 1994

The forensic expert told the court that the likelihood of the sample coming from any man other than the accused was one in three million.

That number describes how rare the profile is. It was offered as the chance he was innocent, the judge repeated it to the jury, and the Court of Appeal quashed the conviction.

TRAFFIC STOPS IN HOUSTON, 2004 TO 2015

A two-dollar roadside kit turns blue for cocaine. It also turns blue for dozens of other things, which has been known since the 1970s.

Blue was taken to mean cocaine. The lab checked only afterwards, and found more than 250 people had already pleaded guilty to possessing nothing.

Both times a number about how often this happens to the innocent was read as the chance of innocence.

R v Deen, Court of Appeal, reported The Times 10 January 1994; rules on presenting DNA statistics were later set in Doheny and Adams [1997] 1 Cr App R 369. Houston: Houston Forensic Science Center and the Harris County conviction integrity unit, reported by ProPublica and The New York Times Magazine, 2016.

A THIRD CAUTIONARY TALE

The detector that cannot tell

WHAT THE VENDORS ADVERTISE

Turnitin has claimed 98%. Copyleaks 99.1%. GPTZero 99%. Winston AI 99.98%.

Every one of those is a statement about machines: if this text came from an AI, we will flag it.

WHAT A FLAGGED STUDENT NEEDS

Of the essays you flag, how many were actually written by an AI?

Vanderbilt ran that number. Even at a 1% false positive rate, their 75,000 submissions a year would falsely accuse 750 students. They switched the tool off in August 2023.

And 1% is generous. Stanford researchers ran seven detectors over 91 TOEFL essays, every one written by a human: 61.3% were flagged as machine-written, and about one in five was flagged by all seven. On essays by native English speakers the same detectors almost never erred.

The same flip, with a transcript at the end of it. A number about how often the tool catches machines, read as the chance that a student cheated.

Liang W, Yuksekgonul M, Mao Y, Wu E, Zou J. GPT detectors are biased against non-native English writers. Patterns 2023. Vanderbilt University disabled Turnitin's AI detection in August 2023. Johns Hopkins and international-student false positives reported by The Markup, 14 August 2023. Vendor accuracy figures are the vendors' own marketing claims.

What we’ve learned

One way a true number can mislead you, and a look ahead.

1 · Probability reversal

The trap: What the test reports gets flipped into what you asked. A positive is read as proof, a negative as the all-clear.

Ask: out of everyone who got this result, how many actually have it?

2 · Next week: Arbitrary categories

The trap: A line drawn across a continuous measurement is read as a fact about your body.

Ask: where do I sit on the curve, not which side of the line?

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Questions, and a conversation

Murray Cantor PhD murray@murraycantor.com

Joel Keenan MD drkeenan@mdvip.com

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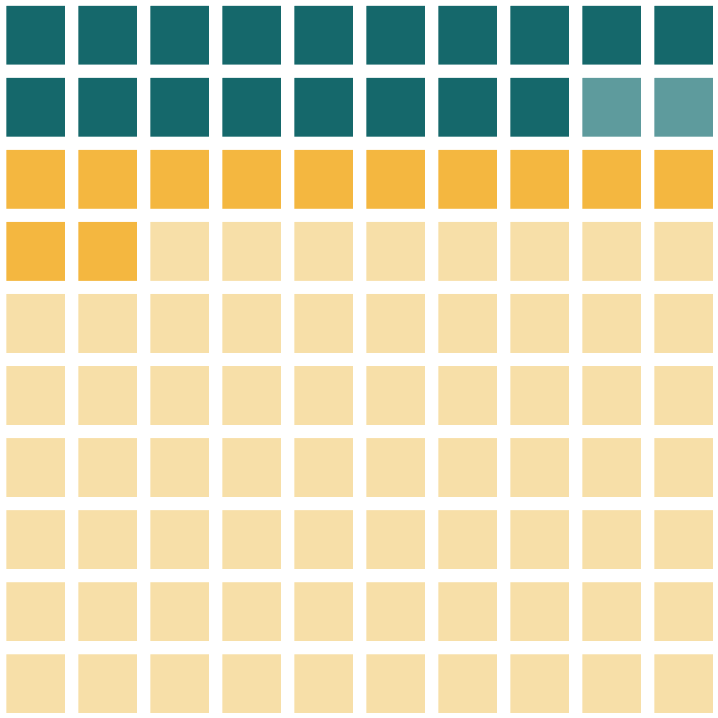
Backup Slides

AN ILLUSTRATIVE RECORD OF 100 MORNINGS

Did it rain overnight, and is the ground wet?



rained and the ground is wet: 18

rained but the ground is dry: 2

no rain but the ground is wet: 12

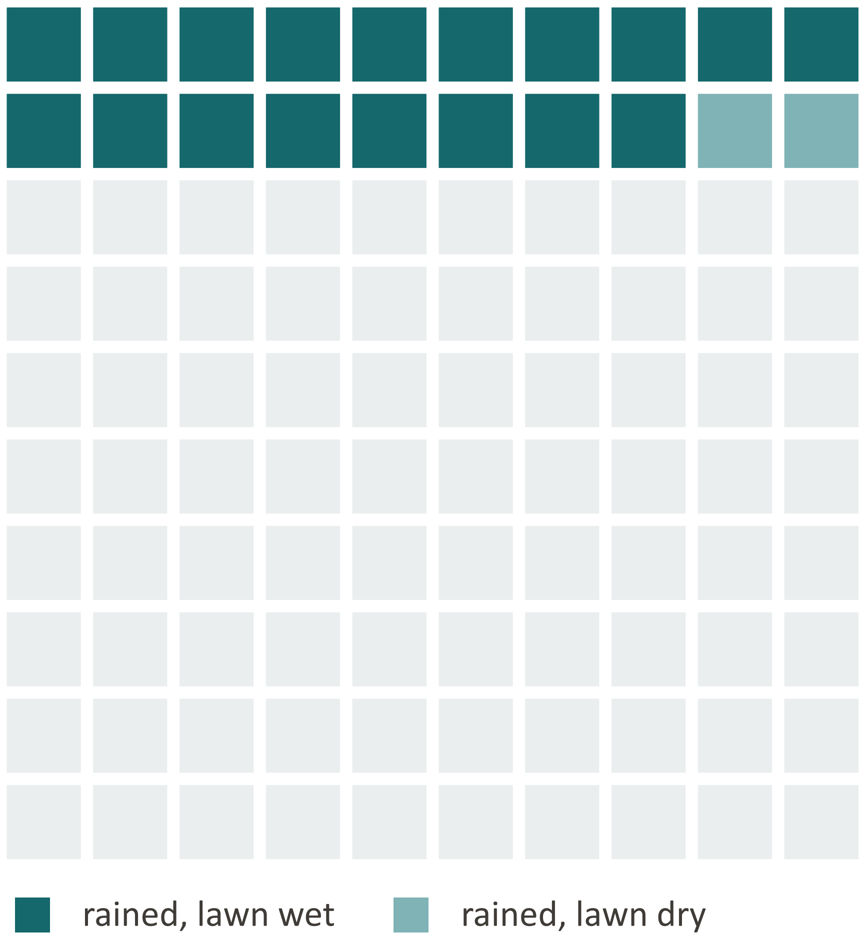
no rain and the ground is dry: 68

It rained on 20 of the 100 mornings, and the ground was wet on 30. Wet without rain comes from sprinklers, dew, or a hose.

Illustrative figures, chosen for a clean count. Not drawn from any real weather record.

FIRST QUESTION

If we know it rained, how likely is a wet lawn?



Look only at the 20 mornings it rained, the highlighted squares. Every dry-night morning is set aside for now.

Of those 20, the mornings the ground is also wet: **18**.

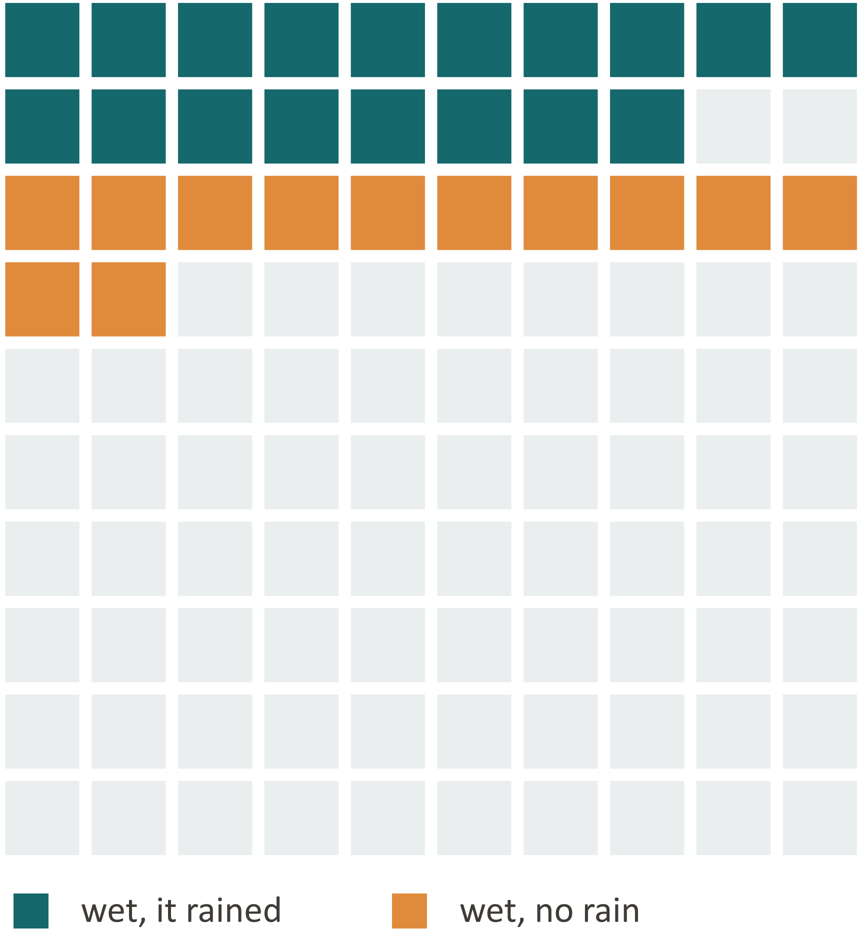
18 of 20 = 90%

Rain almost always leaves the ground wet by morning.

Illustrative figures.

SECOND QUESTION, THE ONE PEOPLE GET BACKWARDS

If we know the lawn is wet, how likely is it that it rained?



Now look only at the 30 mornings the ground is wet, the highlighted squares. Same picture, we just divide by a different group.

Of those 30, the mornings it actually rained: **18**.

18 of 30 = 60%

A wet lawn often means rain, but sprinklers and dew wet it too.

Illustrative figures.

WHAT YOU JUST DID, IN SYMBOLS

Three steps to read P(A | B) off the squares

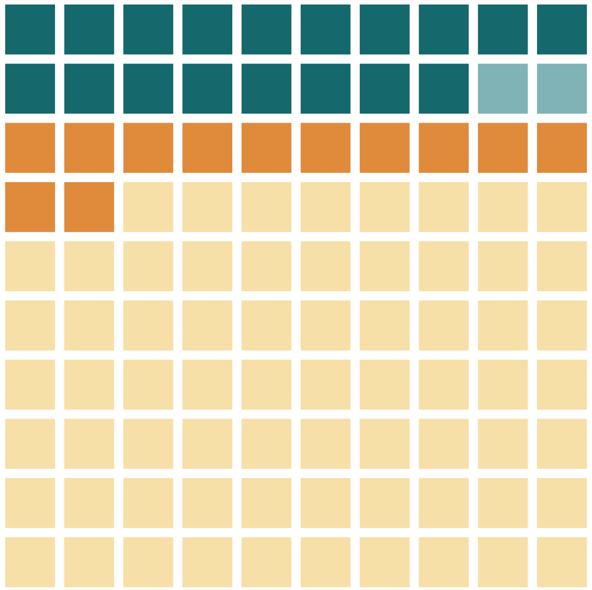
1. **Keep B's group.** Set aside every square outside B, then count what is left. That count is your denominator.

2. **Count the overlap.** Among B's squares, how many are also in A? That count is your numerator.

3. **Divide the two counts.** P(A | B) is just the overlap count over the B count. Nothing more.

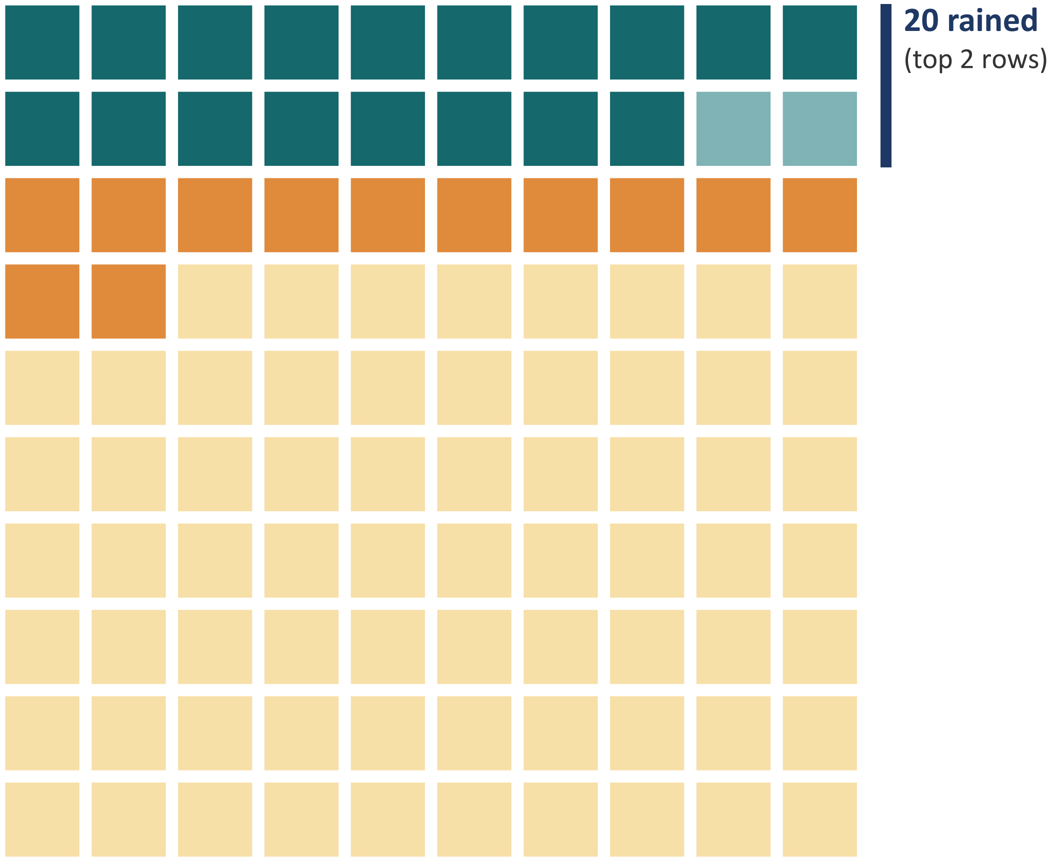
A probability is one count divided by another.

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$



Every morning is one square, so counting squares is all the "probability" you need here.

Same 18 on top, two different worlds beneath



EVERY PROBABILITY IS A RATIO OF COUNTS

If it rained, is it wet?

P(wet | rained) = 18 teal ÷ **20** rained = **90%**

If it is wet, did it rain?

P(rained | wet) = 18 teal ÷ **30** wet = **60%**

Same **18 teal squares** on top both times. Only the count you divide by changes: 20 rainy mornings, or 30 wet ones.

Illustrative figures, carried from the previous slides.

THE BIG IDEA

$P(\text{lawn wet} \mid \text{rain}) = 90\%$. $P(\text{rain} \mid \text{lawn wet}) = 60\%$.

1. **They are different ratios.** Each is a count divided by a count: 18/20 is P(lawn wet | rain), 18/30 is P(rain | lawn wet).

2. **“Rained” is the hypothesis, “wet lawn” the data.** 90% is P(data | hypothesis); 60% is P(hypothesis | data), the one we care about.

3. **A wet lawn has other causes.** Sprinklers and dew dilute it as evidence, so the reverse reads weaker.

4. **This flip is everywhere.** The same backwards reading hides in p-values, medical tests, and courtrooms.

Before reading a wet lawn as proof of rain, *ask what else could have wet it.*

BACKUP: THE SAME FLIP, FOR AN OLDER ROOM

What a callback means after sixty

per 10,000 women aged 60 to 69	specificity 89%	specificity 91%
have cancer	72	72
called back, and do have it	63	63
called back, and do not	1,092	894
total callbacks	1,155	957
P(cancer | called back)	**5.5%**	**6.6%**

The taught slides use 4.4%, the published all-ages figure. Same test, same flip. Prevalence is doing the work.

Inputs. Cancer detection rate 6.3 per 1,000 at ages 60 to 69 (Canadian organised programs report 5.4 to 7.2), sensitivity held at the BCSC 86.9%, so prevalence is about 7.2 per 1,000. Specificity by decade is not published in the BCSC benchmarks: 89% is the all-ages figure, 91% is inferred from recall rates falling with age, so the right-hand column is the softer one.

Corroboration by a different route: Kerlikowske et al. (UCSF series, first screens) report sensitivity 94.1% at ages 60 to 69 and a post-test risk of 0.07 after an abnormal mammogram. Screen-film era, so treat as a check rather than the headline.

THE BIGGER PICTURE

The line has moved, and moved back

2002

Every 1 to 2 years, from age 40.

2009

Every 2 years, from 50. In your forties, an individual decision.

2016

Reaffirmed unchanged.

2024

Every 2 years, from 40 again.

WHY IT MOVED UP IN 2009

In their forties there is less cancer to find, so fewer deaths are averted and more healthy women are called back and biopsied.

WHY IT MOVED BACK IN 2024

Invasive breast cancer in women aged 40 to 49 rose about 2.0% a year from 2015 to 2019. More cancer to find changes the sum.

The mammogram did not change across those twenty-two years. What changed was how much cancer was in the group being screened, and how much weight to give the false positives.

US Preventive Services Task Force breast cancer screening recommendations: 2002 (grade B, from age 40), 16 November 2009 (biennial 50 to 74 grade B; 40 to 49 grade C, an individual decision), 2016 reaffirmation, and 30 April 2024 (biennial 40 to 74, grade B). ACOG, ACS, NCCN and ACR continued to recommend starting at 40 throughout.